

2. Electromagnetic (em) waves are widely used in a variety of ways. Satellites rely on microwaves to communicate with Earth as the microwaves penetrate through the upper layers of our atmosphere. Infra-red waves have a higher frequency than microwaves and are used widely in communications. X-rays and gamma rays are ionising radiation and are both used in medicine; gamma rays can be used in cancer treatment, shrinking tumours. Radio waves are used to transmit TV and radio signals.

(a) The diagram shows an incomplete em spectrum. **Complete the diagram** to show the em spectrum in order. [2]

Radio waves			Visible light	UV light		Gamma rays
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(b) All em waves are transverse. Describe what is meant by a transverse wave. [2]

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(c) Tick (✓) the boxes alongside the **two** correct statements about em waves below. [2]

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| Only radio waves and microwaves transmit information | ☐ |
| All em waves travel at the same speed in a vacuum | ☐ |
| All em waves are a form of radiation | ☐ |
| All em waves have the same frequency | ☐ |
| Radio waves have the smallest wavelength | ☐ |

(d) A student suggests that radio waves are just as harmful to humans as gamma rays. Explain whether or not you agree with this statement. [2]

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